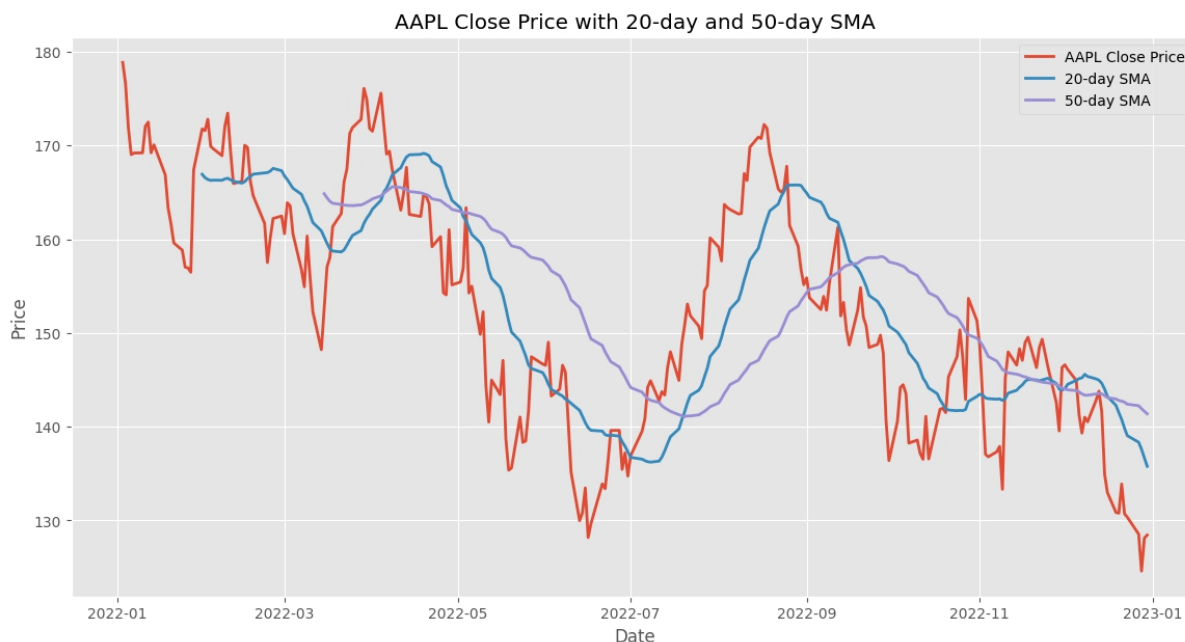


This code snippet calculates two commonly used SMAs: the 20-day (roughly one trading month) and the 50-day (roughly one trading quarter). When the shorter-term SMA crosses above the longer-term SMA, it's often considered a bullish signal, suggesting upward momentum. Conversely, when the shorter-term SMA crosses below the longer-term SMA, it's often seen as a bearish signal.



Exponential Moving Average (EMA)

While the SMA gives equal weight to all prices in the calculation period, the Exponential Moving Average assigns greater weight to more recent prices. This makes the EMA more responsive to new information—like a person who places more emphasis on what happened yesterday than what happened last month.

1. `# Calculate 20-day and 50-day Exponential Moving Averages`
2. `apple['EMA20'] = apple['Close'].ewm(span=20, adjust=False).mean()`
3. `apple['EMA50'] = apple['Close'].ewm(span=50, adjust=False).mean()`
- 4.